A COMPUTATION-FREE PROOF OF THE CHOW RANK OF THE SIX-BY-SIX PERMANENT

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Abstract. We give a self-contained proof that the ordinary Chow rank of the generic six-by-six permanent is thirty-two over every characteristic-zero field. The proof retains the symmetric image-span and factor-span filtration of the earlier manuscript, but removes its computational premises. Small permanent intersections follow from a four-cycle argument. Squarefree symbols are bounded by counting surviving triples, not by enumerating coordinate tables. For the five-dimensional factor span, a squarefree-completion argument pays for missing cubic directions with the middle-rank defect; elementary product bounds suffice in place of all five exact normal-form rank computations. No program output or finite computer-generated table is used in the proof. We make no assertion about border Chow rank.

1. Statement, derivative spaces, and specialization

Work initially over an algebraically closed field $\mathbb{K}$ of characteristic zero. Let $[n] = \{1, \dots, n\}$, with $\binom{a}{b} = 0$ for $0 \le a < b$, and

$$V = \operatorname{span}_{\mathbb{K}}\{x_{ij} : i, j \in [6]\}, \qquad f = \operatorname{perm}_6 = \sum_{\sigma \in S_6} \prod_{i=1}^{6} x_{i,\sigma(i)}.$$

The ordinary Chow rank of a homogeneous degree-m form is the least number of nonzero products of m linear forms whose sum is that form. Nonzero scalar coefficients may be absorbed in one factor. This is also called product rank [2]. We use ordinary decompositions throughout.

Theorem 1.1. *Over every characteristic-zero field,*

$$\operatorname{ChowRank}(\operatorname{perm}_6) = 32.$$

The previous sextic manuscript [3] supplies the framework of a local half-defect bound and two global inequalities. The new arguments below replace its computational endpoints; all other steps are included in full.

For $g \in \operatorname{Sym}^m V$ define

$$C_{a,m-a}(g) : \operatorname{Sym}^a V^* \longrightarrow \operatorname{Sym}^{m-a} V, \quad D \longmapsto D(g), \qquad \mathcal{D}_j(g) = \operatorname{im} C_{m-j,j}(g).$$

Here $0 \le a \le m$, and we set $\mathcal{D}_j(g) = 0$ outside $0 \le j \le m$. Differential operators are composed in characteristic zero. The natural pairing $\langle D, h\rangle = D(h)$ in equal degrees is perfect. In this pairing $C_{a,m-a}(g)$ and $C_{m-a,a}(g)$ are adjoints. The middle catalecticant of a sextic is therefore symmetric. This statement uses the pairing, not an unnormalized monomial-matrix convention.

For a space G of forms, ∂G denotes the span of all first directional derivatives. A form is supported on $L \subset V$ when it belongs to $\operatorname{Sym}^j L$. Its essential-variable space is the least such L. In characteristic zero its dimension is the rank of the first derivative map: vanishing derivatives are exactly the directions annihilating that space. For quadrics this equals the rank of their symmetric matrix.

We will repeatedly use elementary diagonal specialization. A diagonal one-parameter group with distinct weights on a basis takes a subspace, in its Grassmannian, to a coordinate subspace. Indeed the limit exists by projectivity, is fixed by the group, and hence is a sum of the distinct

Date: October 20, 2026.

2020 *Mathematics Subject Classification.* Primary 15A69; Secondary 14N05, 68V05.

Key words and phrases. permanent, Chow rank, product rank, catalecticant, derivative spaces.

one-dimensional eigenspaces. The same group can act on several Grassmannians at once. Maps between the resulting fixed-rank subspace or quotient bundles have rank at the limit at most their rank away from the limit: in local bases the rank conditions are vanishing minors. For an equivariant map the rank along the nonzero orbit is constant. Thus a bound on all coordinate limits is a valid lower bound on the original ranks. For an intersection dimension, viewed as a kernel dimension, the inequality is reversed. We specify the preserving group each time it is used.

2. Permanent structure without enumeration

Put $E_j = \mathcal{D}_j(f)$.

Lemma 2.1. *One has*

$$\dim E_2 = 225, \quad \dim E_3 = 400, \quad \partial E_2 = V, \quad E_2^{(1)} := \{g \in \operatorname{Sym}^3 V : \partial g \subset E_2\} = E_3.$$

Every nonzero element of E_2 has quadratic rank at least four, and every nonzero element of E_3 has at least nine essential variables. Moreover,

$$\dim(E_2 \cap \operatorname{Sym}^2 L) \le \begin{cases} 1, & \dim L \le 5, \\ 3, & \dim L \le 6. \end{cases} \tag{2.1}$$

Proof. The two-by-two and three-by-three subpermanents have disjoint matching supports and form bases of E_2 and E_3. They number $\binom{6}{2}^2 = 225$ and $\binom{6}{3}^2 = 400$. A first derivative of a rectangle permanent gives its opposite variable, so $\partial E_2 = V$. A quadratic belongs to E_2 exactly when squares and same-row or same-column products have coefficient zero, while the two matching coefficients in each rectangle agree.

If every derivative of a cubic g lies in E_2, a cubic monomial with a repeated row or column cannot occur. Differentiate by its remaining factor: the resulting forbidden quadratic has a unique source monomial for this fixed derivative, with nonzero multiplicity. Thus it cannot cancel. The surviving monomials are three-edge matchings. In a fixed three-by-three block, differentiating by the common third factor equates coefficients of matchings differing by a transposition. Transpositions connect all six matchings, so each block contributes one scalar subpermanent. This proves the prolongation identity, including the reverse inclusion.

The row-column torus $x_{ij} \mapsto a_i b_j x_{ij}$ preserves these spaces. Its characters on the individual subpermanents are distinct. A generic one-parameter subgroup, with a scalar normalization, takes any nonzero linear combination to a single subpermanent. Matrix rank and first-derivative rank cannot increase in that limit. A rectangle has rank four. The nine first derivatives of a three-by-three permanent are independent, so its essential-variable dimension is nine. These observations prove the floors.

For the intersection claim choose a row-column subgroup with distinct weights on all thirty-six variables. The limit of an a-plane L is a coordinate a-plane, and the intersection dimension cannot decrease in the limit. A coordinate plane is the edge set of a simple bipartite graph; the supported rectangle permanents count its four-cycles. Two distinct four-cycles use at least six edges. If their union has six edges, they share two adjacent edges, and the union is $K_{2,3}$ or $K_{3,2}$, with exactly three four-cycles. (Two opposite shared edges would determine the same four-cycle.) Hence at most five edges support at most one cycle, and six edges support at most three. No graph enumeration is needed. □

3. Local tools

For a nonzero product $T = \ell_1 \cdots \ell_6$ set

$$L = \operatorname{span}(\ell_1, \dots, \ell_6), \quad \ell = \dim L, \quad U = \mathcal{D}_3(T), \quad u = \dim U, \quad F = \mathcal{D}_2(T), \quad R = F \cap E_2.$$

Third derivatives are spanned by the twenty triple products of factor positions, so $u \le 20$, also for repeated or dependent factors. We never replace U by that possibly larger formal span.

Polarization is the natural map

$$\delta(g) = \sum_j y_j \otimes \partial_{y_j^*} g \in L \otimes F \quad (g \in U),$$

where (y_j) is any basis of L. Given a quotient $P : L \twoheadrightarrow D$ of rank d, and $q_R : F \to F/R$, define

$$\beta = (P \otimes q_R)\delta|_U.$$

The target can also be embedded in $D \otimes (\operatorname{Sym}^2 L/R)$ without changing this rank.

Lemma 3.1. *Let $k = \dim \ker P = \ell - d$. In general,*

$$\operatorname{rank} \beta \geq \max\{0, u - \tbinom{k+2}{3} - d \dim R\}. \tag{3.1}$$

If $\dim R \leq 1$ and every nonzero element of R has rank at least four, then for $k \leq 2$,

$$\ker \beta = U \cap \operatorname{Sym}^3(\ker P), \qquad \operatorname{rank} \beta \geq u - \tbinom{k+2}{3}. \tag{3.2}$$

For the actual $R = F \cap E_2$, the full symbol $d = \ell$ is injective.

Proof. The kernel of $(P \otimes 1)\delta$ on all cubics is $\operatorname{Sym}^3(\ker P)$: in complementary coordinates all the corresponding derivatives must vanish. The inverse image of $D \otimes R$ has dimension at most the kernel dimension plus $d \dim R$, proving (3.1).

For (3.2), a kernel cubic has its complementary derivatives in R. If one is nonzero, changing the complementary coordinates makes these derivatives $q, 0, \ldots, 0$ with $0 \neq q \in R$. Vanishing of the others implies that the cubic, hence q, uses only $\ker P$ and one extra variable. This gives $\operatorname{rank} q \leq k + 1 \leq 3$, a contradiction. Thus all complementary derivatives vanish. The converse and dimension bound follow. If $d = \ell$ and R is actual, a kernel cubic has all derivatives in E_2; it lies in E_3 but is supported on at most six variables. Lemma 2.1 forces it to be zero. □

Lemma 3.2. *The actual middle rank of a sextic product satisfies*

$$\ell = 1, 2, 3, 4, 5 \quad \Longrightarrow \quad u \geq 1, 2, 4, 8, 14, \quad \textit{respectively.}$$

Proof. Choose ℓ independent factors as coordinates. A generic diagonal subgroup takes each remaining factor, after its own scalar normalization, to one coordinate. The product limits to x^α with all $\alpha_i \geq 1$ and $\sum \alpha_i = 6$. Its degree-three derivatives are nonzero multiples of its degree-three monomial divisors. If $r_j = \#\{i : \alpha_i \geq j\}$, the number of these divisors is exactly

$$\binom{\ell}{3} + (\ell - 1)r_2 + r_3, \tag{3.3}$$

by the exponent patterns $1 + 1 + 1$, $2 + 1$, and 3. For $\ell = 1$ this is one. For $\ell = 2$, either $r_2 \geq 2$, or the sole large exponent is five and $r_2 = r_3 = 1$; the count is at least two. For $\ell = 3$, if $r_2 = 1$ that exponent is four, giving four divisors; if $r_2 \geq 2$ there are at least five. For $\ell = 4$, $r_2 = 1$ forces an exponent three, giving eight divisors, whereas $r_2 \geq 2$ gives at least ten. For $\ell = 5$ exactly one exponent is two, giving $10 + 4 = 14$. Middle rank cannot increase in the specialization, so these floors apply to T. This is a formula and an elementary inequality, not a partition table. □

4. Squarefree symbols by counting triples

For $n = 5$ or 6, let $F_{\mathrm{sf}} \subset \operatorname{Sym}^2 \mathbb{K}^n$ and $U_{\mathrm{sf}} \subset \operatorname{Sym}^3 \mathbb{K}^n$ be the spans of squarefree pair and triple monomials in $z_1, \ldots, z_n$. A triple I has polarization $\sum_{i \in I} z_i \otimes z_{I \setminus \{i\}}$.

For a coordinate kernel indexed by $B \subset [n]$, write $O = [n] \setminus B$, $|O| = d$. A coordinate quadratic quotient kills an edge set $A \subset \binom{[n]}{2}$. Distinct source triples have disjoint vertex-edge output coordinates. Thus the rank is the number of surviving columns. A column is killed precisely when every $i \in I \cap O$ has $I \setminus \{i\} \in A$. Equivalently, the killed-column count is

$$\binom{|B|}{3} + d\, e_A(B) + \sum_{b \in B} \binom{\deg_A(b, O)}{2} + \#\{\text{triangles in } A[O]\}. \tag{4.1}$$

Here $e_A(B)$ counts edges contained in B, $\deg_A(b, O)$ counts the neighbors of b in O, and $A[O]$ is the induced graph. The four terms count triples with zero, one, two, and three vertices in O. For arbitrary subspaces $\ker P$ and $R \subset F_{\mathrm{sf}}$, simultaneously specialize by the variable diagonal torus, with distinct variable and pair weights. The limits are the coordinate objects above. Polarization is natural, so the ranks are constant along the nonzero orbit and can only drop at the limit. The ensuing coordinate lower bounds therefore hold for all such subspaces, even when the limiting quadrics lose rank.

Lemma 4.1. *For $n = 5$, $\dim R \le 1$, and $d = 1, 2$, the squarefree symbol rank is at least $5, 7$, respectively.*

Proof. At a coordinate limit at most one edge is killed. In (4.1) the last two terms vanish, so the rank is at least $10 - \binom{5-d}{3} - d$. This is five at $d = 1$ and seven at $d = 2$. The same bounds hold if $R = 0$. These are two evaluations of the derived formula, not an enumeration of coordinate pairs. □

Lemma 4.2. *For $n = 6$ and $\dim R \le 3$, the squarefree symbol has rank at least $10d/3$ when $0 \le d \le 5$.*

Proof. It suffices to treat a coordinate edge set with at most three edges. If $0 \le d \le 3$, every killed triple not contained in B has at least one factorization as an outside vertex and an edge of A. There are at most $d|A| \le 3d$ such factorizations. Hence the rank is at least $20 - \binom{6-d}{3} - 3d$. Its excess over $10d/3$ is

$$20 - \binom{6-d}{3} - \frac{19d}{3} = \frac{d(3-d)(12-d)}{6} \ge 0.$$

If $d = 4$, B has two vertices. When its internal edge is killed, it accounts for four triples, leaving at most two further edges. These can kill at most one two-outside triple and cannot form a triangle, so at most five triples are killed. When the internal edge is absent, the cross-edge term in (4.1) is at most $\binom{3}{2} = 3$; a triangle would use all three edges and give only one killed triple instead. Thus the rank is at least fifteen, larger than $40/3$. If $d = 5$, B has one vertex and no internal edge. The same cross-edge or triangle argument kills at most three triples, leaving rank at least seventeen, larger than $50/3$. We deliberately do not claim a rank-twenty bound for arbitrary R at $d = 6$: three killed edges can form a triangle. Full injectivity for the actual permanent intersection will be supplied by Lemma 3.1. □

5. The half-defect estimate without normal-form ranks

Proposition 5.1. *For every sextic Chow term and every quotient of its actual factor span,*

$$\boxed{\operatorname{rank} \beta + \frac{20-u}{2} \ge \frac{10}{3} d.} \tag{5.1}$$

Proof. For $d = 0$ the claim follows from $u \le 20$.

Factor span at most four. Here $\dim R \le 1$ by (2.1). If $d = 1$, the constant $c = \binom{\ell+1}{3} + 1$ in (3.1) is at most eleven. For every real u,

$$\max(0, u - c) + \frac{20-u}{2} = 10 - \frac{c}{2} + \frac{|u-c|}{2} \ge \frac{9}{2} > \frac{10}{3}.$$

If $d = 2$, the complementary dimension is at most two, and (3.2) with $u \ge 2$ gives $\operatorname{rank} \beta + (20-u)/2 \ge 10 + u/2 - 4 \ge 7 > 20/3$. If $d = 3$, the complementary dimension is at most one and $u \ge 4$, giving at least $10 + u/2 - 1 \ge 11 > 10$. For $d = 4$, necessarily $\ell = 4$, $u \ge 8$, and the same bound with complementary dimension zero is at least $14 > 40/3$.

Six independent factors. Now $u = 20$, the actual spaces are squarefree, and $\dim R \le 3$. Lemma 4.2 proves the assertion for $d \le 5$. For $d = 6$, Lemma 3.1 gives $\operatorname{rank} \beta = 20 = 10d/3$.

Five-dimensional factor span. Here $14 \le u \le 20$ and $\dim R \le 1$. For $d = 3, 4, 5$, (3.2) gives, respectively, lower bounds $13, 16, 17$ for $\operatorname{rank} \beta + (20-u)/2$, all at least $10d/3$. Only $d = 1, 2$

remain. Choosing five independent factors, permuting and rescaling them puts the product, up to a nonzero scalar, in a form

$$T_s = x_1x_2x_3x_4x_5(x_1 + \cdots + x_s), \qquad 1 \le s \le 5. \tag{5.2}$$

We do not compute its exact derivative ranks. Instead we construct a squarefree cubic subspace $U_0 \subset U$ with

$$m := 10 - \dim U_0 \le \frac{20-u}{2}. \tag{5.3}$$

For $s = 1$, the product is $x_1^2x_2x_3x_4x_5$ and all ten squarefree cubics are degree-three divisors, hence actual derivatives. Take $m = 0$. For $s \ge 2$, the explicit identity

$$\partial_j^2\partial_k T_s = 2 \prod_{i\in[5]\setminus\{j,k\}} x_i \quad (j \le s,\ k \ne j)$$

puts in U every squarefree triple whose complementary pair meets $[s]$. For $s = 4, 5$ this includes all ten, so again $m = 0$. For $s = 3$ it includes nine (all except $x_1x_2x_3$), so $m = 1$. For $s = 2$ it includes seven (all except $x_1x_2x_j$, $j = 3, 4, 5$), so $m = 3$.

It remains to verify the budget for $s = 2, 3$. If A, B involve disjoint variable sets, ordinary differentiation and bidegrees give

$$\mathcal{D}_j(AB) = \bigoplus_{a+b=j} \mathcal{D}_a(A) \otimes \mathcal{D}_b(B). \tag{5.4}$$

Indeed pure derivatives in the two variable sets generate each indicated product space, and distinct bidegrees do not overlap. For $s = 2$, take $A = x_1x_2(x_1 + x_2)$ and $B = x_3x_4x_5$. The degree-$0, 1, 2, 3$ derivative dimensions of A are bounded by $1, 2, 2, 1$ (the degree-two space consists of first derivatives in two variables); for B they are $1, 3, 3, 1$. Thus

$$u \le 1 \cdot 1 + 2 \cdot 3 + 2 \cdot 3 + 1 \cdot 1 = 14,$$

so $(20 - u)/2 \ge 3 = m$. For $s = 3$, take $A = x_1x_2x_3(x_1 + x_2 + x_3)$ and $B = x_4x_5$. The degree-$1, 2, 3$ spaces of A have dimensions at most $3, 6, 3$; the degree-$2, 1, 0$ spaces of B have dimensions $1, 2, 1$. Hence

$$u \le 3 \cdot 1 + 6 \cdot 2 + 3 \cdot 1 = 18,$$

so $(20 - u)/2 \ge 1 = m$. This proves (5.3) in every form.

To use it, extend polarization to all ten squarefree cubics on these five coordinates, with target $D \otimes (\operatorname{Sym}^2 L/R)$. On the squarefree quadratic space the effective killed subspace is $R \cap F_{\mathrm{sf}}$, of dimension at most one. Lemma 4.1 bounds the extended rank by at least $a_1 = 5$ or $a_2 = 7$. Restricting its source to U_0 loses at most its codimension m. On U_0 it is the actual symbol followed by the injective target inclusion $F/R \hookrightarrow \operatorname{Sym}^2 L/R$. Therefore

$$\operatorname{rank}\beta + \frac{20-u}{2} \ge a_d - m + \frac{20-u}{2} \ge a_d \ge \frac{10d}{3}, \qquad d = 1, 2.$$

This is a comparison with a squarefree completion, not an identification of the actual derivative space with a formal factor-product span. All factor-span dimensions and all quotients are covered. □

6. Global cancellation and the lower bound

Lemma 6.1 (Symmetric image-span inequality)**.** *Let $A_i : W^* \to W$ be symmetric linear maps of ranks u_i, and put $D = \dim \sum_i \operatorname{im} A_i$. Then*

$$\operatorname{rank}\left(\sum_i A_i\right) \ge 2D - \sum_i u_i.$$

Proof. Choose injections $B_i : \mathbb{K}^{u_i} \hookrightarrow W$ with images $\operatorname{im} A_i$. Symmetry implies $\ker A_i = (\operatorname{im} A_i)^{\perp}$, so $A_i = B_i J_i B_i^*$ for invertible symmetric maps $J_i : (\mathbb{K}^{u_i})^* \to \mathbb{K}^{u_i}$. Let B be their horizontal concatenation and $J = \bigoplus_i J_i$. Then $\operatorname{rank} B = \operatorname{rank} B^* = D$ and $\sum_i A_i = BJB^*$. The dimension of the intermediate space is $\sum_i u_i$; the rank of a composition is at least the sum of the two ranks minus that dimension. It follows that $\operatorname{rank}(BJB^*) \geq 2D - \sum_i u_i$. □

Suppose $f = \sum_{i=1}^{N} T_i$ is an actual Chow decomposition. Write

$$U_i = \mathcal{D}_3(T_i), \quad u_i = \dim U_i, \quad \Delta = \sum_i (20 - u_i) \geq 0, \quad U = \sum_i U_i, \quad \dim U = 400 + h.$$

Since $E_3 \subset U$, $h \geq 0$. Apply Lemma 6.1 to the symmetric maps $A_i = C_{3,3}(T_i)$. Their sum is $C_{3,3}(f)$ of rank four hundred. Consequently

$$h \leq 10N - 200 - \frac{\Delta}{2}. \tag{6.1}$$

Set $F_i = \mathcal{D}_2(T_i)$, $F = \sum_i F_i$, and $Q = F/E_2$. Linearity of catalecticants gives $E_2 \subset F$. With $q : F \to Q$ define

$$b : U \longrightarrow V \otimes Q, \qquad b(g) = (1_V \otimes q)\delta(g).$$

This is well defined because $\partial U \subset F$. Its kernel is $U \cap E_2^{(1)} = E_3$, by Lemma 2.1. Thus $\operatorname{rank} b = h$.

Let L_i be the actual factor span of T_i, order the terms arbitrarily, and put $W_i = L_1 + \cdots + L_i$, $W_0 = 0$, $d_i = \dim(W_i/W_{i-1})$. Since $V = \partial E_2 \subset \partial F \subset \sum_i L_i$, one has $\sum_i d_i = 36$. Put $Z_i = b(U_i) \subset L_i \otimes Q$ and $H_i = Z_1 + \cdots + Z_i$. Projection of $W_i \otimes Q$ to $(W_i/W_{i-1}) \otimes Q$ kills H_{i-1}, so

$$\dim H_i - \dim H_{i-1} \geq \operatorname{rank}\big((P_i \otimes q|_{F_i})\delta|_{U_i}\big),$$

where $P_i : L_i \twoheadrightarrow L_i/(L_i \cap W_{i-1})$ has rank d_i. The map $F_i/(F_i \cap E_2) \hookrightarrow F/E_2$ is injective, so the right side is precisely the local symbol rank from Proposition 5.1. This argument does not assume independence of the unprojected images Z_i. Since $H_N = b(U)$, summation gives

$$h \geq \sum_i \left(\frac{10 d_i}{3} - \frac{20 - u_i}{2} \right) = 120 - \frac{\Delta}{2}. \tag{6.2}$$

The same defect appears with the same coefficient in (6.1) and (6.2). Cancelling it yields $120 \leq 10N - 200$, hence

$$\operatorname{ChowRank}(f) \geq 32.$$

7. The upper bound, field scope, and proof boundary

Glynn's identity [1] gives

$$\operatorname{perm}_6 = \frac{1}{32} \sum_{\substack{\epsilon \in \{1,-1\}^6 \\ \epsilon_1 = 1}} \left(\prod_{i=1}^{6} \epsilon_i \right) \prod_{j=1}^{6} \left(\sum_{i=1}^{6} \epsilon_i x_{ij} \right). \tag{7.1}$$

For completeness, expanding a summand chooses one row in each column. If row i is used m_i times, the sign sum for $i = 2, \ldots, 6$ is $\sum_{\epsilon_i = \pm 1} \epsilon_i^{m_i + 1}$. It is zero unless each of $m_2, \ldots, m_6$ is odd. These are then at least one, and $\sum m_i = 6$ forces all six counts to equal one: increasing any of $m_2, \ldots, m_6$ to the next odd value would already require degree seven. The coefficient of a permutation monomial is $32/32 = 1$. Thus (7.1) is the asserted thirty-two-term decomposition. For any characteristic-zero field, a hypothetical shorter decomposition would remain one after extending to its algebraic closure, where the lower bound applies. The upper bound is defined over $\mathbb{Q}$. This proves Theorem 1.1 over every such field.

Remark 7.1 (What is, and is not, used). The four-cycle argument replaces the graph scan. Formula (4.1) and its inequalities replace the squarefree rank table. Formula (3.3) replaces partition minima. In the dependent five-dimensional case only elementary upper bounds $u \leq 14, 18$ and explicit squarefree derivatives are used; neither the five exact middle ranks

nor the five exact quadratic ranks are proof premises. Finally the absolute-value inequality and the small-complement bounds replace all finite interval minimizations. No ancillary file is included by this source. Optional independent exact checks accompany the revision but no lemma cites their output. The proof concerns ordinary rank; it does not establish border rank or a general-size equality.

Provenance and disclosure

This is an AI-assisted revision of the author's October 16 sextic manuscript. ChatGPT assisted with developing the replacement arguments, internal auditing, and preparation of the text and optional checks. Internal review and exact regression checks are not external peer review or formal proof-assistant certification. The frozen earlier release is retained separately.

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